GRADE 12 GEOGRAPHY

EUEE MASTER STUDY GUIDE

Ethiopian University Entrance Exam
Complete Exam Preparation

Plate Tectonics · Climate Change · Resources · Population · Development

Simplified · Exam-Focused · High-Score Strategies

UNIT 1

MAJOR GEOLOGICAL PROCESSES — PLATE TECTONICS

1.1 Continental Drift Theory

Definition — Continental Drift

Alfred Wegener's (1912) theory that all continents were once joined in a supercontinent called Pangaea, which broke apart and drifted to present positions.

Evidence for continental drift:

- **Jigsaw fit:** coastlines of South America and Africa fit together
- **Fossil evidence:** identical fossils (Mesosaurus, Glossopteris) found on now-separate continents
- **Rock matching:** identical rock formations on opposite sides of Atlantic
- **Palaeoclimate evidence:** coal (tropical) found in Antarctica; glacial deposits in Africa

Limitation: Wegener could not explain the mechanism of movement.

1.2 Plate Tectonics Theory

Definition — Plate Tectonics

The theory that Earth's lithosphere is divided into large plates that move on the semi-molten asthenosphere. Convection currents in the mantle drive the movement.

Evidence:

- Sea-floor spreading (Hess, 1960s): new ocean floor formed at mid-ocean ridges
- Magnetic striping on the ocean floor
- Deep-sea trenches where plates subduct

1.3 Plate Boundaries

Divergent (constructive)	Plates move apart	Mid-ocean ridges, rift valleys, volcanoes	Mid-Atlantic Ridge, East African Rift

Convergent (destructive)	Plates move together	Trenches, mountain ranges, volcanoes	Andes, Himalayas, Mariana Trench
Transform (conservative)	Plates slide past each other	Earthquakes, no volcanoes	San Andreas Fault
Collision	Two continental plates collide	Fold mountains	Himalayas (India + Eurasia)

1.4 Major Geological Processes

Earthquakes:

- **Focus (hypocentre):** point inside Earth where earthquake originates
- **Epicentre:** point on surface directly above focus
- **Seismic waves:** P-waves (primary, fastest, travel through solids/liquids), S-waves (secondary, only solids), Surface waves (most destructive)
- **Richter scale:** logarithmic scale measuring earthquake magnitude

Volcanoes:

- **Shield volcanoes:** gentle slopes; basaltic lava; non-explosive (Hawaii)
- **Composite/Stratovolcanoes:** steep slopes; explosive; andesitic lava (Mt. Etna, Vesuvius)
- **Cinder cone:** small, steep, made of tephra; most common type

Ethiopian context:

- Ethiopia sits on the East African Rift System — active tectonic zone
- Erta Ale (Afar) — active lava lake volcano

EXAM FOCUS

- Know all three plate boundary types, features, and examples.
- East African Rift Valley — convergent or divergent? DIVERGENT — Africa splitting apart.
- P-waves vs S-waves: P travels through liquids too; S does not.

PRACTICE QUESTIONS

1. Explain the evidence Wegener used for continental drift.
2. Compare divergent and convergent plate boundaries.
3. What is the difference between the focus and epicentre of an earthquake?
4. Describe the features of a shield volcano vs a composite volcano.

UNIT 2

CLIMATE CHANGE

2.1 Basic Concepts

Definition — Climate

The average weather conditions of an area over a long period (30+ years). Different from weather (short-term atmospheric conditions).

Definition — Climate Change

Long-term shifts in temperatures and weather patterns. Since the 1800s, human activities — especially burning fossil fuels — have been the main driver.

Greenhouse Effect:

- Solar radiation passes through atmosphere and warms Earth
- Earth radiates heat (infrared); greenhouse gases trap some of it
- Natural greenhouse effect keeps Earth ~33°C warmer than it would be
- Enhanced greenhouse effect (human-caused) = global warming
- **GHGs:** CO_2 (dominant), CH_4, N_2O, water vapour, CFCs

2.3 Natural vs Human-Induced Climate Change

Natural causes:

- Milankovitch cycles (changes in Earth's orbit and tilt)
- Solar activity variations, volcanic eruptions, El Niño/La Niña

Human causes:

- Burning fossil fuels (main cause of rising CO_2)
- Deforestation (reduces CO_2 absorption)
- Agriculture (methane from livestock and paddy fields, N_2O from fertilisers)
- Urbanisation and industrial emissions

2.4 Consequences of Climate Change

- Rising global temperatures (1.1°C above pre-industrial levels)
- Melting polar ice → sea level rise → coastal flooding
- More frequent extreme events: drought, floods, hurricanes
- Biodiversity loss and ecosystem disruption

- Food insecurity — reduced crop yields in tropical regions
- **Ethiopia specifically:** more frequent droughts, shifting rainfall patterns, highland temperature rise

2.5-2.7 Responses and International Agreements

- **Mitigation:** reducing GHG emissions (renewable energy, energy efficiency, reforestation)
- **Adaptation:** adjusting to climate impacts (drought-resistant crops, sea walls, early warning systems)
- **UNFCCC (1992):** international framework for climate action
- **Kyoto Protocol (1997):** binding targets for developed countries to reduce emissions
- **Paris Agreement (2015):** all countries commit to keep warming below 2°C (target 1.5°C)
- **Ethiopia's CRGE Strategy:** Climate Resilient Green Economy — develop without high emissions

UNIT 3

MANAGEMENT OF CONFLICT OVER RESOURCES

3.1 Sustainable Development

Definition — Sustainable Development

Development that meets present needs without compromising future generations' ability to meet theirs. (Brundtland Report, 1987)

- **Three pillars:** Economic sustainability + Social equity + Environmental protection
- **SDGs (2015):** 17 Sustainable Development Goals; target 2030; replace MDGs

3.2 Resource Use Conflicts

Types of resource conflicts:

- **Water conflicts:** Nile water dispute (Ethiopia, Egypt, Sudan over GERD — Grand Ethiopian Renaissance Dam)
- **Land conflicts:** between farmers and pastoralists; land grabbing
- **Forest conflicts:** deforestation for agriculture vs conservation
- **Mineral conflicts:** resource curse — oil/mineral wealth fuelling conflict (DRC, South Sudan)

3.3-3.4 Governance and Indigenous Conflict Resolution

- **Good governance principles:** transparency, accountability, rule of law, participation, equity
- **Indigenous conflict resolution (Ethiopia):** Gada system (Oromo — democratic governance), Shimgelina (elder mediation), Xeer (Somali customary law)
- **GERD:** 3.2 billion USD project; 6,450 MW capacity; disputes with Egypt over Nile water

UNIT 4

POPULATION POLICIES, PROGRAMMES & THE ENVIRONMENT

4.1 Population Growth Theories

Definition — Demographic Transition Model

A model showing how birth rates and death rates change as a country develops. Population growth is highest in Stage 2.

Stage 1 (Pre-industrial)	High	High	Stable (low)
Stage 2 (Early industrial)	High	Falling	Rapid growth
Stage 3 (Late industrial)	Falling	Low	Slower growth
Stage 4 (Post-industrial)	Low	Low	Stable (high)

Malthus (1798): population grows geometrically; food arithmetically → famine inevitable. Critics: underestimated technology.

4.2-4.3 Population Policies

- **Pro-natalist policies:** encourage more births (France, Russia) — baby bonuses, tax breaks, maternity leave
- **Anti-natalist policies:** reduce birth rate (China's One Child Policy 1979–2015; India's family planning)

Ethiopia's population:

- ~120 million (2023); second most populous in Africa
- High TFR historically; government promotion of family planning
- Health Extension Programme (HEP) — trained health workers in every village

4.5 Population and Environment

- Rapid population growth → more land cleared → deforestation, soil erosion
- Overconsumption in rich countries has greater environmental impact than population growth in poor countries
- **IPAT equation:** Impact = Population × Affluence × Technology

EXAM FOCUS

- Demographic Transition Model — all 4 stages, characteristics.
- GERD and Nile water dispute — Ethiopia perspective.
- Indigenous conflict resolution systems of Ethiopia.

LAST-MINUTE REVISION GUIDE

Pangaea	**Wegener's supercontinent; broke apart ~200 MYA**
Divergent boundary	**Plates apart; ridges, rift valleys; East African Rift**
Convergent boundary	**Plates together; trenches, mountains; Himalayas, Andes**
Transform boundary	**Plates slide; earthquakes; San Andreas Fault**
P-waves	**Primary; fastest; travel through solid AND liquid**
S-waves	**Secondary; only through solid; help locate liquid outer core**
Greenhouse gases	**CO_2, CH_4, N_2O, water vapour, CFCs**
Paris Agreement	**2015; keep warming below 2°C (target 1.5°C)**
Kyoto Protocol	**1997; binding emissions cuts for developed nations**
CRGE Ethiopia	**Climate Resilient Green Economy strategy**
DTM Stage 2	**High birth rate + falling death rate = fastest population growth**
GERD	**Grand Ethiopian Renaissance Dam; 6,450 MW; Nile dispute with Egypt**
Gada System	**Oromo indigenous governance and conflict resolution**
Sustainable Dev.	**Meet present needs without compromising future generations**

EUEE EXAM STRATEGY

Time Management

- **Skim first (5 min):** Read all questions. Mark easy (E), medium (M), hard (H).
- **Easy first:** Answer all E questions to bank safe marks and build confidence.
- **Pace yourself:** Aim for ~2 minutes per question. If stuck over 3 minutes, skip and return.
- **Show working:** Even wrong answers earn partial marks if steps are correct.
- **Final review:** Use last 10 minutes to check calculations, names, and dates.

Answering Techniques

- Multiple-choice: eliminate obviously wrong answers first, then choose from what remains.
- Definition questions: always include the key term in your answer.
- Explanation questions: structure as Definition → Example → Importance.
- Diagram questions: label all parts, use arrows correctly, add units where needed.
- Essay questions: write a clear intro, 3-4 main points with evidence, strong conclusion.

Common Exam Traps

- Confusing similar terms — always define the exact term asked.
- Incomplete answers — re-read every question before finalising.
- Time overrun — if a question takes too long, note it and move on.
- Skipping diagrams — a well-labelled diagram often earns more marks than a paragraph.
- Not reading all options in MCQ — always read all four choices before selecting.